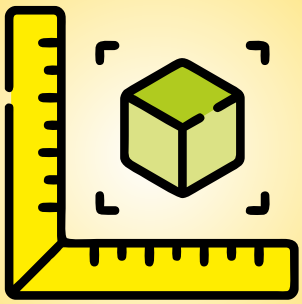
ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)
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PHYSICS

JEE (MAIN+ADVANCED)

NURTURE COURSE



Exercise

Unit & Dimension, Basic Maths and Vectors

(English Medium)

EXERCISE (O)

SINGLE CORRECT TYPE QUESTIONS

Units & Dimensions

1. The dimensional formula for which of the following pair is not the same
(A) impulse and momentum (B) torque and work
(C) stress and pressure (D) momentum and angular momentum

UD0030

2. Which of the following can be a set of fundamental quantities
(A) length, velocity, time (B) momentum, mass, velocity
(C) force, mass, velocity (D) momentum, time, frequency

UD0031

3. Which of the following functions of A and B may be performed if A and B possess different dimensions?

(A) $\frac{A}{B}$ (B) A + B (C) A – B (D) None

UD0032

4. The velocity v of a particle at time t is given by $v = at + \frac{b}{t+c}$, where a, b and c are constants. The dimensions of a, b and c are respectively :–
(A) LT^{-2}, L and T (B) L^2, T and LT^2 (C) LT^2, LT and L (D) L, LT and T^2

UD0033

5. If area (A), velocity (v), and density (ρ) are base units, then the dimensional formula of force can be represented as :-
(A) $Av\rho$ (B) $Av^2\rho$ (C) $Av\rho^2$ (D) $A^2v\rho$

UD0034

6. Density of wood is 0.5 g/ cc in the CGS system of units. The corresponding value in MKS units is :-
(A) 500 (B) 5 (C) 0.5 (D) 5000

UD0035

7. In a book, the answer for a particular question is expressed as $b = \frac{ma}{k}\left[\sqrt{1+\frac{2k\ell}{ma}}\right]$ here m represents mass, a represents acceleration, ℓ represents length. The unit of b should be :-
(A) m/s (B) m/s^2 (C) meter (D) /sec

UD0036

8. The frequency f of vibrations of a mass m suspended from a spring of spring constant k is given by $f=Cm^xk^y$, where C is a dimensionless constant. The values of x and y are, respectively

(A) $\frac{1}{2}, \frac{1}{2}$ (B) $-\frac{1}{2}, -\frac{1}{2}$ (C) $\frac{1}{2}, -\frac{1}{2}$ (D) $-\frac{1}{2}, \frac{1}{2}$

UD0037

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9. If force, acceleration and time are taken as fundamental quantities, then the dimensions of length will be:
(A) FT^2 (B) $F^{-1}A^2T^{-1}$ (C) FA^2T (D) AT^2

UD0038

10. In a particular system the units of length, mass and time are chosen to be 10 cm, 10 g and 0.1 s respectively. The unit of force in this system will be equal to :-
(A) 0.1 N (B) 1 N (C) 10 N (D) 100 N

UD0039

11. The units of three physical quantities x, y and z are $gcm^2 s^{-5}$, gs^{-1} and cms^{-2} respectively. The relation among the units of x, y and z is :-
(A) $z = x^2y$ (B) $y^2 = xz$ (C) $x = yz^2$ (D) $x = y^2z$

UD0040

12. In a certain system of units, 1 unit of time is 5 sec, 1 unit of mass is 20 kg and 1 unit of length is 10 m. In this system, one unit of power will correspond to :-
(A) 16 watts (B) $\frac{1}{16}$ watts (C) 25 watts (D) none of these

UD0076

13. If the unit of length be doubled then the numerical value of the universal gravitation constant G will become (with respect to present value)
(A) Double (B) Half (C) 8 times (D) 1/8 times

UD0077

14. In two different systems of units an acceleration is represented by the same number, while a velocity is represented by numbers in the ratio 1 : 3. The ratios of unit of length and time are respectively
(A) 3, 9 (B) 9, 3 (C) 1, 1 (D) None of these

UD0079

15. The equation of the stationary wave is $y = 2A\sin\left(\frac{2\pi ct}{\lambda}\right)\cos\left(\frac{2\pi x}{\lambda}\right)$ which of the following statement is wrong?
(A) The unit of ct is same as that of λ. (B) The unit of x is same as that of λ.
(C) The unit of $\frac{2\pi c}{\lambda}$ is same as that of $\frac{2\pi x}{\lambda t}$ (D) The unit of $\frac{c}{\lambda}$ is same as that of $\frac{x}{\lambda}$

UD0043

Basic maths

16. The angle subtended by the moon's diameter at a point on the earth is about 0.50°. Use this and the fact that the moon is about 384000 km away to find the approximate diameter of the moon.



(A) 192000 km (B) 3350 km (C) 1600 km (D) 1920 km

UD0041

17. The approximate value of x where x = sin 1° cos 1°, is
(A) $\frac{\pi}{90}$ (B) 2 (C) $\frac{\pi}{180}$ (D) $\frac{\pi}{45}$

UD0144

18. Using small angle approximation find value of (tan 3°)(cos 4°) :-
(A) $\frac{\pi}{3}$ (B) $\frac{\pi}{60}$ (C) $\frac{\pi}{45}$ (D) $\frac{\pi}{90}$

UD0145

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Addition of vectors

19. A man moves towards 3 m north then 4 m towards east and finally 5 m towards 37° south of west. His displacement from origin is :-

(A) $5\sqrt{2}$m (B) 0m (C) 1 m (D) 12 m

UD0055

20. If the resultant of two forces of magnitudes P and Q acting at a point at an angle of 60° is $\sqrt{7}$ Q, then P/Q is :-

(A) 1 (B) $\frac{3}{2}$ (C) 2 (D) 4 **UD0048**

21. The resultant of two forces, one double the other in magnitude is perpendicular to the smaller of the two forces. The angle between two forces is :-

(A) 150° (B) 90° (C) 60° (D) 120°

UD0046

22. There are two force vectors, one of 5N and other of 12N at what angle the two vectors be added to get resultant vector of 17N, 7N and 13N respectively.

(A) 0°, 180° and 90° (B) 0°, 90° and 180°
(C) 0°, 90° and 90° (D) 180°, 0° and 90° **UD0049**

23. A body placed in free space, is simultaneously acted upon by three forces $\vec{F}_1$, $\vec{F}_2$ and $\vec{F}_3$. The body is in equilibrium and the forces $\vec{F}_1$ and $\vec{F}_2$ are known to be 36 N due north and 27 N due east respectively. Which of the following best describes the force $\vec{F}_3$?

(A) 36 N due south. (B) 53 N due 60° south of east
(C) 45 N due 53° south of west (D) 45 N due 37° north of west

UD0050

24. The ratio of maximum and minimum magnitudes of the resultant of two vectors $\vec{A}$ and $\vec{B}$ is 3 : 2. The relation between A and B is

(A) A = 5B (B) A = B (C) A = 3B (D) A = 4B

UD0051

25. Two vectors $\vec{a}$ and $\vec{b}$ add to give a resultant $\vec{c} = \vec{a} + \vec{b}$. In which of these cases angle between $\vec{a}$ and $\vec{b}$ is maximum: (a, b, c represent the magnitudes of respective vectors)

(A) c = a + b (B) $c^2 = a^2 + b^2$ (C) c = a – b (D) can not be determined

UD0053

26. Three concurrent forces of the same magnitude are in equilibrium. What is the angle between the force? Also name the triangle formed by the force as sides :-

(A) 60° equilateral triangle (B) 120° equilateral triangle
(C) 120°, 30°, 30° an isosceles triangle (D) 120° an obtuse angled triangle

UD0045

Resolution of vectors and unit vector

27. The x and y components of a force are 2N and –3N. The force is :

(A) $2\hat{i} - 3\hat{j}$ (B) $2\hat{i} + 3\hat{j}$ (C) $-2\hat{i} - 3\hat{j}$ (D) $3\hat{i} + 2\hat{j}$

UD0146

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28. **Statement 1** : Unit vector has a unit though its magnitude is one
and
Statement 2 : Unit vector is obtained by dividing a vector by its own magnitude.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true. **UD0083**

29. Due to some unknown interaction, force F experienced by a particle is given by the following equation.

$$\vec{F} = -\frac{A}{r^3}\vec{r}$$

Where A is positive constant and r distance of the particle from origin of a coordinate system. Dimensions of constant A are :-
(A) ML^2T^{-2} (B) ML^3T^{-2} (C) ML^4T^{-2} (D) ML^0T^0
UD0044

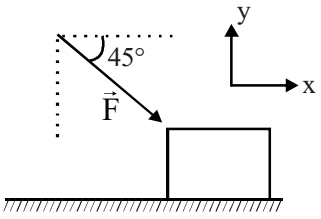
30. A bird moves from point (1, –2) to (4, 2). If the speed of the bird is 10 m/s, then the velocity vector of the bird is
(A) $5\left(\hat{i}-2\hat{j}\right)$ (B) $5\left(4\hat{i}+2\hat{j}\right)$ (C) $0.6i+0.8\hat{j}$ (D) $6\hat{i}+8\hat{j}$
UD0057

31. A man rows a boat with a speed of 18 km/hr in north–west direction. The shoreline makes an angle of 15° south of west. Obtain the component of the velocity of the boat along the shoreline.
(A) 9 km/hr (B) $18\frac{\sqrt{3}}{2}$ km/hr (C) 18 cos (15°) km/hr (D) 18 cos (75°) km/hr
UD0082

32. For the given vector $\vec{A} = 3\hat{i} - 4\hat{j} + 10\hat{k}$, the ratio of magnitude of its component on the x-y plane and the component on z–axis is :-
(A) 2 (B) $\frac{1}{2}$ (C) 1 (D) None
UD0060

33. The angles which $\hat{i}+\hat{j}+\sqrt{2}\hat{k}$ makes with x-axis, y-axis and z-axis respectively are :
(A) 60º, 60º and 45º (B) 60º, 45º and 45º (C) 45º, 45º and 45º (D) 60º, 60º and 60º
UD0147

34. A person pushes a box kept on a horizontal surface with force of 100 N. In unit vector notation force $\vec{F}$ can be expressed as :



(A) $100\ (\hat{i}+\hat{j})$ (B) $100\,(\hat{i}-\hat{j})$ (C) $50\sqrt{2}\ (\hat{i}-\hat{j})$ (D) $50\sqrt{2}\ (\hat{i}+\hat{j})$
UD0059

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35. A vector of magnitude 10 m in the direction 37° south of west has its initial point at (5 m, 2 m). If positive x–axis represents the east and positive y–axis the north, the coordinates of its terminal point are

(A) (–3 m, –4 m) (B) (3 m, 4 m) (C) (–4 m, 6 m) (D) (–4 m,–6 m)

UD0084

36. A body moves in anticlockwise direction on a circular path in the x-y plane. The radius of the circular path is 5 m and its centre is at the origin. In a certain interval of time, displacement of the body is observed to be 6 m in the positive y-direction. Which of the following is true?

(A) Its initial position vector is $5\hat{i}$ m. (B) Its initial position vector is $(-3\hat{i}+4\hat{j})$ m.

(C) Its final position vector is $\left(4\hat{i}+3\hat{j}\right)$ m. (D) Its final position vector is $6\hat{j}$ m.

UD0086

37. A plumber steps down 1 m out of his truck and walks 50 m east and then 25 m south, and then takes an elevator to the basement of the building 9 m below street level. If the east, the north and the upward direction are represented by the positive x, y and z–axes, which one of the following represents displacement (meters) of the plumber?

(A) $50\hat{i}-25\hat{j}-9\hat{k}$ (B) $50\hat{i}+25\hat{j}-9\hat{k}$ (C) $50\hat{i}-25\hat{j}-10\hat{k}$ (D) $50\hat{i}+25\hat{j}-10\hat{k}$

UD0085

Scalar product of vectors

38. In a methane (CH_4) molecule each hydrogen atom is at a corner of a regular tetrahedron with the carbon atom at the centre. In coordinates where one of theC–H bonds is in the direction of $\hat{\mathrm{i}}+\hat{\mathrm{j}}+\hat{\mathrm{k}}$, an adjacent C–H bond in the $\hat{\mathrm{i}}-\hat{\mathrm{j}}-\hat{\mathrm{k}}$ direction. Then angle between these two bonds :-

(A) $\cos^{-1}\left(-\frac{2}{3}\right)$ (B) $\cos^{-1}\left(\frac{2}{3}\right)$ (C) $\cos^{-1}\left(-\frac{1}{3}\right)$ (D) $\cos^{-1}\left(\frac{1}{3}\right)$

UD0062

39. If $\vec{a}$ and $\vec{b}$ are two unit vectors such that $\vec{a}+2\vec{b}$ and $5\vec{a}-4\vec{b}$ are perpendicular to each other then the angle between $\vec{a}$ and $\vec{b}$ is :-

(A) 45° (B) 60° (C) $\cos^{-1}\left(\frac{1}{3}\right)$ (D) $\cos^{-1}\left(\frac{2}{7}\right)$

UD0063

40. A particle moves from a position $3\hat{\mathrm{i}}+2\hat{\mathrm{j}}-6\hat{\mathrm{k}}$ to a position $14\hat{\mathrm{i}}+13\hat{\mathrm{j}}+9\hat{\mathrm{k}}$ in m and a uniform force of $4\hat{\mathrm{i}}+\hat{\mathrm{j}}+3\hat{\mathrm{k}}$ N acts on it. The work done by the force is :-

(A) 200 J (B) 100 J (C) 300 J (D) 500 J

UD0065

41. Which of the following is perpendicular to $\hat{i}-\hat{j}-\hat{k}$?

(A) $\hat{i}+\hat{j}+\hat{k}$ (B) $-\hat{i}+\hat{j}+\hat{k}$ (C) $\hat{i}+\hat{j}-\hat{k}$ (D) none of these

UD0066

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MULTIPLE CORRECT TYPE QUESTIONS

42. For the equation x= AC sin (Bt) + D e $^{(BCt)}$, where x and t represent position and time respectively, which of the following is/are **CORRECT** :-

(A) Dimension of AC is LT^{-1} (B) Dimension of B is T^{-1}

(C) Dimension of AC and D are same (D) Dimension of C is T^{-1}

UD0069

43. Four forces acting on a particle keep it in equilibrium, then :-

(A) the force must be coplanar.

(B) the forces cannot be coplanar.

(C) the forces may or may not be coplanar.

(D) if three of these forces are coplanar, so must be the fourth.

UD0093

44. The four pairs of force vectors are given, which pairs of force vectors cannot be added to give a resultant vector of magnitude 10 N?

(A) 2N, 13 N (B) 5N, 16 N (C) 7N, 8N (D) 100N, 105 N

UD0097

COMPREHENSION TYPE QUESTIONS

Paragraph for Question Nos. 45 and 46

For any particle moving with some velocity $(\vec{v})$ & acceleration $(\vec{a})$, tangential acceleration & normal acceleration are defined as follows

Tangential acceleration – The component of acceleration in the direction of velocity.

Normal acceleration – The component of acceleration in the direction perpendicular to velocity.

If at a given instant, velocity & acceleration of a particle are given by.

45. Find the tangential acceleration of the particle at the given instant :-

(A) $17\left(4\hat{i}+3\hat{j}\right)$ (B) $\frac{17}{5}\left(4\hat{i}+3\hat{j}\right)$ (C) $17\left(4\hat{i}-3\hat{j}\right)$ (D) $\frac{17}{5}\left(4\hat{i}-3\hat{j}\right)$

UD0071

46. Find the normal acceleration of the particle at the given instant :-

(A) $\frac{-9\hat{i}+12\hat{j}+50\hat{k}}{5}$ (B) $\frac{9\hat{i}-12\hat{j}-50\hat{k}}{5}$

(C) $\frac{-18\hat{i}+24\hat{j}+100\hat{k}}{5}$ (D) $\frac{18\hat{i}-24\hat{j}-100\hat{k}}{5}$

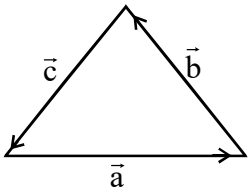
UD0071

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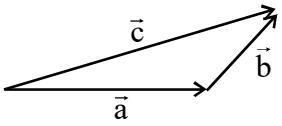
MATRIX MATCH TYPE QUESTION

47. Column-I show vector diagram relating three vectors $\vec{a}, \vec{b}$ and $\vec{c}$. Match the vector equation in column-II, with vector diagram in column-I :

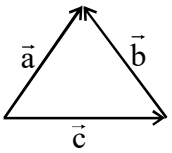
Column-I **Column-II**

(A) 

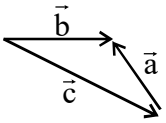
(P) $\vec{a}-(\vec{b}+\vec{c})=0$

(B) 

(Q) $\vec{b}-\vec{c}=\vec{a}$

(C) 

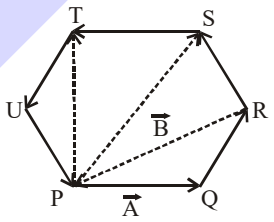
(R) $\vec{a}+\vec{b}=-\vec{c}$

(D) 

(S) $\vec{a}+\vec{b}=\vec{c}$

UD0073

48. In a regular hexagon two vectors $\overrightarrow{PQ}=\vec{A}$, $\overrightarrow{RP}=\vec{B}$. Express other vector's in term of them :-



Column–I **Column–II**

(A) $\overrightarrow{PS}$

(B) $\overrightarrow{PT}$

(C) $\overrightarrow{RS}$

(D) $\overrightarrow{TS}$

(P) $-2\vec{B}-3\vec{A}$

(Q) $-\vec{B}-\vec{A}$

(R) $-\vec{B}-2\vec{A}$

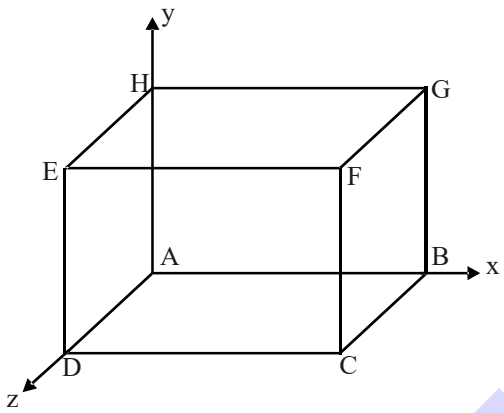
(S) $-2(\vec{B}+\vec{A})$

(T) $\vec{A}$

UD0074

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49. Figure shows a cube of edge length a.



	Column-I		**Column-II**
(A)	The angle between AF and x-axis	(P)	$60°$
(B)	Angle between AF and DG	(Q)	$\cos^{-1}\frac{1}{3}$
(C)	Angle between AE and AG	(R)	$\cos^{-1}\frac{1}{\sqrt{3}}$
		(S)	$\cos^{-1}\sqrt{\frac{2}{3}}$

UD0107

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EXERCISE (S)

Units & Dimensions

1. A particle is in a unidirectional potential field where the potential energy (U) of a particle depends on the x-coordinate given by $U_x = k(1-\cos ax)$ & k and 'a' are constants. Find the physical dimensions of 'a' & k.

UD0001

2. The equation for the speed of sound in a gas states that $v = \sqrt{\gamma k_B T / m}$. Speed v is measured in m/s, γ is a dimensionless constant, T is temperature in kelvin (K), and m is mass in kg. Find the SI unit for the Boltzmann constant, k_B?

UD0002

3. The time period (T) of a spring mass system depends upon mass (m) & spring constant (k) & length of the spring (ℓ) $\left[k = \frac{Force}{length}\right]$. Find the relation among T, m, ℓ & k using dimensional method.

UD0003

4. The distance moved by a particle in time t from centre of a ring under the influence of its gravity is given by $x = a\sin\omega t$, where a & ω are constants. If ω is found to depend on the radius of the ring (r), its mass (m) and universal gravitational constant (G). Using dimensional analysis find an expression for ω in terms of r, m and G.

UD0004

5. Assume that the largest stone of mass 'm' that can be moved by a flowing river depends upon the velocity of flow v, the density d & the acceleration due to gravity g . If 'm' varies as the Kth power of the velocity of flow, then find the value of K .

UD0006

6. Given $\vec{F} = \frac{\vec{a}}{t}$ where symbols have their usual meaning. The dimensions of a is . **UD0007**

Basic Maths

7. Find the value of :
(i) $\sin 74°$ (ii) $\cos 106°$ (iii) $\sin 15°$ (iv) $\cos 75°$

UD0148

8. Quito, a city in Ecuador and Kampala, a city situated in Uganda both lie on the Equator. The longitude of Quito is 82°30' W and that of Kampala is 37°30' E. What is the distance from Quito to Kampala.
(a) along the shortest surface path
(b) along a direct (through-the-Earth) path? (The radius of the Earth is 6.4×10^6 m)

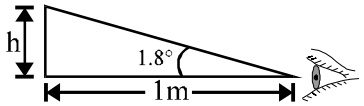
UD0020

9. Use the approximation $(1+x)^n \approx 1+nx, |x| << 1$, to find approximate value for

(a) $\sqrt{99}$ (b) $\frac{1}{1.01}$ **UD0021**

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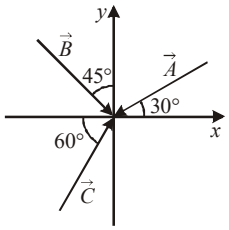
10. A normal human eye can see an object making an angle of 1.8° at the eye. What is the approximate height of object which can be seen by an eye placed at a distance of 1 m from the object.



UD0149

Addition of vectors

11. Vectors $\vec{A}$, $\vec{B}$ and $\vec{C}$ are shown in figure. Find angle between.



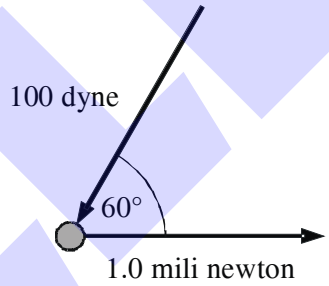
(i) $\vec{A}$ and $\vec{B}$ (ii) $\vec{A}$ and $\vec{C}$ (iii) $\vec{B}$ and $\vec{C}$

UD0150

12. A block is applied two forces of magnitude 5N each. One force is acting towards East and the other acting along 60° North of East. The resultant of the two forces (in N) is of magnitude :-

UD0008

13. Two forces act on a particle simultaneously as shown in the figure. Find net force in milli newton on the particle. [Dyne is the CGS unit of force]

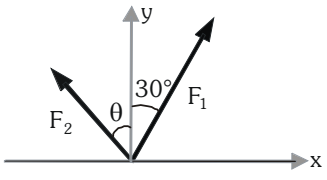


UD0009

14. The maximum and minimum magnitudes of the resultant of two forces are 35 N and 5 N respectively. Find the magnitude of resultant force when act orthogonally to each other.

UD0010

15. In the figure, F_1 and F_2, the two unknown forces give a resultant force of $80\sqrt{3}$ N along the y–axis. It is required that F_2 must have minimum magnitude. Find the magnitudes of F_1 and F_2.



UD0028

16. Three boys are pushing horizontally a box placed on horizontal table. One is pushing towards north with a $5\sqrt{3}$N force. The second is pushing towards east and third pushes with a force 10 N such that the box is in equilibrium. Find the magnitude of the force, second boy is applying in newton.

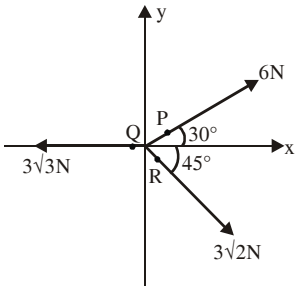
UD0015

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17. Vector B has x, y and z components of 4.00, 6.00 and 3.00 units, respectively. Calculate the magnitude of B and the angles that B makes with the coordinates axes.

UD0013

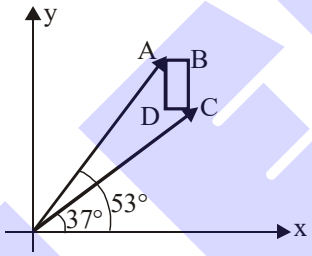
18. Three ants P, Q and R are pulling a grain with forces of magnitude 6N, $3\sqrt{3}$N and $3\sqrt{2}$N as shown in the figure. Find the magnitude of resultant force (in N) acting on the grain.



UD0014

Resolution of vectors and unit vector

19. The farm house shown in figure has rectangular shape and has sides parallel to the chosen x and y axes. The position vector of corner A is 125 m at 53° and corner C is 100 m at 37° from x axis. Find the length of the fencing required in meter.



UD0012

Scalar product of vectors

20. (a) Calculate $\vec{r} = \vec{a} - \vec{b} + \vec{c}$ where $\vec{a} = 5\hat{i} + 4\hat{j} - 6\hat{k}$, $\vec{b} = -2\hat{i} + 2\hat{j} + 3\hat{k}$ and $\vec{c} = 4\hat{i} + 3\hat{j} + 2\hat{k}$.

(b) Calculate the angle between $\vec{r}$ and the z-axis.

(c) Find the angle between $\vec{a}$ and $\vec{b}$

UD0018

21. A particle is displaced from A ≡ (2, 2, 4) to B ≡ (5, –3, –1). A constant force of 34 N acts in the direction of $\vec{AP}$, where P ≡ (10, 2, –11). (Coordinates are in m).

(i) Find the $\vec{F}$. (ii) Find the work done by the force to cause the displacement.

UD0029

22. If the velocity of a particle is $(2\hat{i} + 3\hat{j} - 4\hat{k})$ and its acceleration is $(-\hat{i} + 2\hat{j} + \hat{k})$ and angle between them is $\frac{n\pi}{4}$. The value of n is.

UD0019

23. Consider the two vectors : $\vec{L} = 1\hat{i} + 2\hat{j} + 3\hat{k}$ and $\vec{l} = 4\hat{i} + 5\hat{j} + 6\hat{k}$. Find the value of the scalar α such that the vector $\vec{L} - \alpha\vec{l}$ is perpendicular to $\vec{L}$.

UD0016

24. Find components of vector $\vec{a} = \hat{i} + \hat{j} + 3\hat{k}$ in directions parallel to and perpendicular to vector $\vec{b} = \hat{i} + \hat{j}$.

UD0017

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EXERCISE (JM)

1. An ideal gas enclosed in a vertical cylindrical container supports a freely moving piston of mass M. The piston and the cylinder have equal cross sectional area A. When the piston is in equilibrium, the volume of the gas is V_0 and its pressure is P_0. The piston is slightly displaced from the equilibrium position and released. Assuming that the system is completely isolated from its surrounding, the piston executes a simple harmonic motion with frequency. (γ is dimensionless constant)

[JEE Main-2013]

(1) $\frac{1}{2\pi}\frac{A\gamma P_0}{V_0 M}$ (2) $\frac{1}{2\pi}\frac{V_0 M P_0}{A^2\gamma}$ (3) $\frac{1}{2\pi}\sqrt{\frac{A^2\gamma P_0}{MV_0}}$ (4) $\frac{1}{2\pi}\sqrt{\frac{MV_0}{A\gamma P_0}}$

UD0108

SELECTED PROBLEMS FROM JEE-MAINS ONLINE PAPERS

2. Expression for time in terms of G (universal gravitational constant), h (Planck constant) and c (speed of light) is proportional to : **[JEE-Main-2019_Jan]**

(1) $\sqrt{\frac{Gh}{c^3}}$ (2) $\sqrt{\frac{hc^5}{G}}$ (3) $\sqrt{\frac{c^3}{Gh}}$ (4) $\sqrt{\frac{Gh}{c^5}}$

UD0151

3. The density of a material in SI units is 128 kg m^{-3}. In certain units in which the unit of length is 25 cm and the unit of mass is 50 g, the numerical value of density of the material is :

[JEE-Main-2019_Jan]

(1) 410 (2) 640 (3) 16 (4) 40

UD0152

4. If speed (V), acceleration (A) and force (F) are considered as fundamental units, the dimension of Young's modulus will be :- **[JEE-Main-2019_Jan]**

(1) $V^{-2}A^2F^2$ (2) $V^{-4}A^2F$ (3) $V^{-4}A^{-2}F$ (4) $V^{-2}A^2F^{-2}$

UD0153

5. The force of interaction between two atoms is given by $F = \alpha\beta\exp\left(-\frac{x^2}{\alpha kT}\right)$; where x is the distance, k is the Boltzmann constant and T is temperature and α and β are two constants. The dimension of β is: (Dimension of Boltzmann constant is $ML^2T^{-2}K^{-1}$) **[JEE-Main-2019_Jan]**

(1) $M^2L^2T^{-2}$ (2) M^2LT^{-4} (3) $M^0L^2T^{-4}$ (4) MLT^{-2}

UD0154

6. Two forces P and Q of magnitude 2F and 3F, respectively, are at an angle θ with each other. If the force Q is doubled, then their resultant also gets doubled. Then, the angle is :

[JEE-Main-2019_Jan]

(1) 30° (2) 60° (3) 90° (4) 120°

UD0155

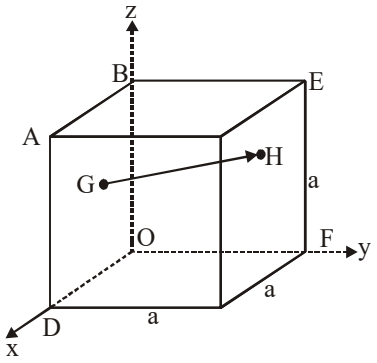
7. Two vectors $\vec{A}$ and $\vec{B}$ have equal magnitudes. The magnitude of $(\vec{A}+\vec{B})$ is 'n' times the magnitude of $(\vec{A}-\vec{B})$. The angle between $\vec{A}$ and $\vec{B}$ is : **[JEE-Main-2019_Jan]**

(1) $\sin^{-1}\left[\frac{n^2-1}{n^2+1}\right]$ (2) $\cos^{-1}\left[\frac{n-1}{n+1}\right]$ (3) $\cos^{-1}\left[\frac{n^2-1}{n^2+1}\right]$ (4) $\sin^{-1}\left[\frac{n-1}{n+1}\right]$

UD0156

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8. In the cube of side 'a' shown in the figure, the vector from the central point of the face ABOD to the central point of the face BEFO will be: **[JEE-Main-2019_Jan]**



(1) $\frac{1}{2}a\left(\hat{i}-\hat{k}\right)$ (2) $\frac{1}{2}a\left(\hat{j}-\hat{i}\right)$ (3) $\frac{1}{2}a\left(\hat{k}-\hat{i}\right)$ (4) $\frac{1}{2}a\left(\hat{j}-\hat{k}\right)$

UD0157

9. Let $\left|\vec{A}_1\right|=3$, $\left|\vec{A}_2\right|=5$ and $\left|\vec{A}_1+\vec{A}_2\right|=5$. The value of $\left(2\vec{A}_1+3\vec{A}_2\right).\left(3\vec{A}_1-2\vec{A}_2\right)$ is :- **[JEE-Main-2019_April]**

(1) –112.5 (2) –106.5 (3) –118.5 (4) –99.5

UD0158

10. A quantity f is given by $f=\sqrt{\frac{hc^5}{G}}$ where c is speed of light, G universal gravitational constant and h is the Planck's constant. Dimension of f is that of : **[JEE-Main-2020_Jan]**

(1) Momentum (2) Area (3) Energy (4) Volume

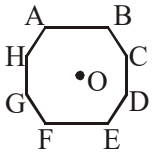
UD0159

11. Amount of solar energy received on the earth's surface per unit area per unit time is defined a solar constant. Dimension of solar constant is: **[JEE-Main-2020_Sep]**

(1) ML^2T^{-2} (2) MLT^{-2} (3) $M^2L^0T^{-1}$ (4) ML^0T^{-3}

UD0160

12. In an octagon ABCDEFGH of equal side, what is the sum of $\overrightarrow{AB}+\overrightarrow{AC}+\overrightarrow{AD}+\overrightarrow{AE}+\overrightarrow{AF}+\overrightarrow{AG}+\overrightarrow{AH}$' if, $\overrightarrow{AO}=2\hat{i}+3\hat{j}-4\hat{k}$? **[JEE-Main-2021_Feb]**



(1) $-16\hat{i}-24\hat{j}+32\hat{k}$ (2) $16\hat{i}+24\hat{j}-32\hat{k}$ (3) $16\hat{i}+24\hat{j}+32\hat{k}$ (4) $16\hat{i}-24\hat{j}+32\hat{k}$

UD0161

13. If time (t), velocity (v), and angular momentum (ℓ) are taken as the fundamental units. Then the dimension of mass (m) in terms of t, v and ℓ is : **[JEE-Main-2021_July]**

(1) $[t^{-1}\ v^1\ \ell^{-2}]$ (2) $[t^1\ v^2\ \ell^{-1}]$ (3) $[t^{-2}\ v^{-1}\ \ell^1]$ (4) $[t^{-1}\ v^{-2}\ \ell^1]$

UD0162

14. The force is given in terms of time t and displacement x by the equation
F = A cos Bx + C sin Dt

The dimensional formula of $\frac{AD}{B}$ is : **[JEE-Main-2021_July]**

(1) $[M^0\ L\ T^{-1}]$ (2) $[M\ L^2\ T^{-3}]$ (3) $[M^1\ L^1\ T^{-2}]$ (4) $[M^2\ L^2\ T^{-3}]$

UD0163

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15. What will be the projection of vector $\vec{A} = \hat{i} + \hat{j} + \hat{k}$ on vector $\vec{B} = \hat{i} + \hat{j}$? **[JEE-Main-2021_July]**

(1) $\sqrt{2}(\hat{i} + \hat{j} + \hat{k})$ (2) $2(\hat{i} + \hat{j} + \hat{k})$ (3) $\sqrt{2}(\hat{i} + \hat{j})$ (4) $(\hat{i} + \hat{j})$

UD0164

16. If force (F), length (L) and time (T) are taken as the fundamental quantities. Then what will be the dimension of density : **[JEE-Main-2021_Aug]**

(1) $[FL^{-4}T^2]$ (2) $[FL^{-3}T^2]$ (3) $[FL^{-5}T^2]$ (4) $[FL^{-3}T^3]$

UD0165

17. Match List-I with List-II. **[JEE-Main-2021_Aug]**

	List-I		**List-II**
(a)	Torque	(i)	MLT^{-1}
(b)	Impulse	(ii)	MT^{-2}
(c)	Tension	(iii)	ML^2T^{-2}
(d)	Surface Tension	(iv)	MLT^{-2}

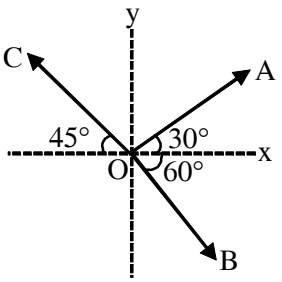
Choose the most appropriate answer from the option given below :

(1) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii) (2) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

(3) (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii) (4) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

UD0166

18. The magnitude of vectors $\overrightarrow{OA}$, $\overrightarrow{OB}$ and $\overrightarrow{OC}$ in the given figure are equal. The direction of $\overrightarrow{OA} + \overrightarrow{OB} - \overrightarrow{OC}$ with x-axis will be :- **[JEE-Main-2021_Aug]**

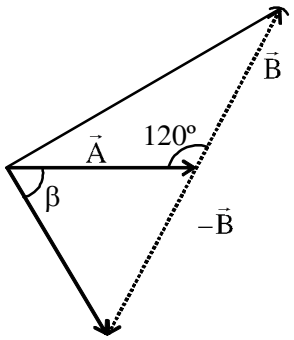


(1) $\tan^{-1}\dfrac{(1-\sqrt{3}-\sqrt{2})}{(1+\sqrt{3}+\sqrt{2})}$ (2) $\tan^{-1}\dfrac{(\sqrt{3}-1+\sqrt{2})}{(1+\sqrt{3}-\sqrt{2})}$

(3) $\tan^{-1}\dfrac{(\sqrt{3}-1+\sqrt{2})}{(1-\sqrt{3}+\sqrt{2})}$ (4) $\tan^{-1}\dfrac{(1+\sqrt{3}-\sqrt{2})}{(1-\sqrt{3}-\sqrt{2})}$

UD0167

19. The angle between vector $(\vec{A})$ and $(\vec{A} - \vec{B})$ is : **[JEE-Main-2021_Aug]**



(1) $\tan^{-1}\left(\dfrac{-\frac{B}{2}}{A - B\frac{\sqrt{3}}{2}}\right)$ (2) $\tan^{-1}\left(\dfrac{A}{0.7B}\right)$ (3) $\tan^{-1}\left(\dfrac{\sqrt{3}B}{2A - B}\right)$ (4) $\tan^{-1}\left(\dfrac{B\cos\theta}{A - B\sin\theta}\right)$

UD0168

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\1-U&D, Basic Maths & Vector, NUM, K-1D,2D\Eng\Eng\01_Unit & Dimension, Basic Maths and Vector\01_Eng.p65

20. Which of the following relations is true for two unit vectors $\hat{A}$ and $\hat{B}$ making an angle θ to each other?

[JEE-Main-2022_June]

(A) $|\hat{A}+\hat{B}|=|\hat{A}-\hat{B}|\tan\frac{\theta}{2}$

(B) $|\hat{A}-\hat{B}|=|\hat{A}+\hat{B}|\tan\frac{\theta}{2}$

(C) $|\hat{A}+\hat{B}|=|\hat{A}-\hat{B}|\cos\frac{\theta}{2}$

(D) $|\hat{A}-\hat{B}|=|\hat{A}+\hat{B}|\cos\frac{\theta}{2}$

UD0169

21. An expression for a dimensionless quantity P is given by $P=\frac{\alpha}{\beta}\log_e\left(\frac{kt}{\beta x}\right)$; where α and β are constants, x is distance ; k is Boltzmann constant and t is the temperature. Then the dimensions of α will be :

[JEE-Main-2022_June]

(A) $[M^0L^{-1}T^0]$ (B) $[ML^0T^{-2}]$ (C) $[MLT^{-2}]$ (D) $[ML^2T^{-2}]$

UD0170

22. In Vander Waals equation $\left[P+\frac{a}{V^2}\right][V-b]=RT$; P is pressure, V is volume, R is universal gas constant and T is temperature. The ratio of constants $\frac{a}{b}$ is dimensionally equal to : **[JEE-Main-2022_June]**

(A) $\frac{P}{V}$ (B) $\frac{V}{P}$ (C) PV (D) PV^3

UD0171

23. Match List I with List II.

	List I		**List II**
(A)	Torque	(I)	Nms^{-1}
(B)	Stress	(II)	$J\ kg^{-1}$
(C)	Latent Heat	(III)	Nm
(D)	Power	(IV)	Nm^{-2}

Choose the correct answer from the options given below: **[JEE-Main-2022_July]**

(A) A-III, B-II, C-I, D-IV (B) A-III, B-IV, C-II, D-I

(C) A-IV, B-I, C-III, D-II (D) A-II, B-III, C-I, D-IV

UD0172

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EXERCISE (J-A)

1. Match List I with List II and select the correct answer using the codes given below the lists : **[JEE Advanced-2013]**

	List I		**List II**
P.	Boltzmann constant	1.	$[ML^2T^{-1}]$
Q.	Coefficient of viscosity	2.	$[ML^{-1}T^{-1}]$
R.	Planck constant	3.	$[MLT^{-3}K^{-1}]$
S.	Thermal conductivity	4.	$[ML^2T^{-2}K^{-1}]$

Codes :

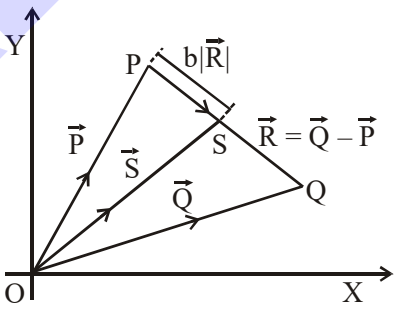
	P	Q	R	S
(A)	3	1	2	4
(B)	3	2	1	4
(C)	4	2	1	3
(D)	4	1	2	3

UD0109

2. To find the distance d over which a signal can be seen clearly in foggy conditions, a railways engineer uses dimensional analysis and assumes that the distance depends on the mass density ρ of the fog, intensity (power/area) S of the light from the signal and its frequency f. The engineer finds that d is proportional to $S^{1/n}$. The value of n is. **[JEE Advanced-2014]**

UD0110

3. Three vectors $\vec{P},\vec{Q}$ and $\vec{R}$ are shown in the figure. Let S be any point on the vector $\vec{R}$. The distance between the points P and S is $b|\vec{R}|$. The general relation among vectors $\vec{P},\vec{Q}$ and $\vec{S}$ is : **[JEE Advanced-2017]**



(A) $\vec{S}=(1-b)\vec{P}+b^2\vec{Q}$

(B) $\vec{S}=(b-1)\vec{P}+b\vec{Q}$

(C) $\vec{S}=(1-b)\vec{P}+b\vec{Q}$

(D) $\vec{S}=(1-b^2)\vec{P}+b\vec{Q}$

UD0112

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\1-U&D, Basic Maths & Vector, NUM, K-1D,2D\Eng\Eng\01_Unit & Dimension, Basic Maths and Vector\01_Eng.p65

4. Two vectors $\vec{A}$ and $\vec{B}$ are defined as $\vec{A} = a\hat{i}$ and $\vec{B} = a\left(\cos\omega t\hat{i} + \sin\omega t\hat{j}\right)$, whre a is a constant and $\omega = \pi/6$ rad s^{-1}. If $|\vec{A} + \vec{B}| = \sqrt{3}|\vec{A} - \vec{B}|$ at time $t = \tau$ for the first time, the value of τ, in seconds, is _____. **[JEE Advanced-2018]**

UD0113

5. Let us consider a system of units in which mass and angular momentum are dimensionless. If length has dimension of L, which of the following statement (s) is/are correct ?

[JEE Advanced-2019]

(1) The dimension of force is L^{-3}

(2) The dimension of energy is L^{-2}

(3) The dimension of power is L^{-5}

(4) The dimension of linear momentum is L^{-1}

UD0115

6. Sometimes it is convenient to construct a system of units so that all quantities can be expressed in terms of only one physical quantity. In one such system, dimensions of different quantities are given in terms of a quantity X as follows: [position] = $[X^{\alpha}]$; [speed] = $[X^{\beta}]$; [acceleration] = $[X^{p}]$; [linear momentum] = $[X^{q}]$; [force] = $[X^{r}]$. Then -

[JEE Advanced-2020]

(A) $\alpha + p = 2\beta$ (B) $p + q - r = \beta$ (C) $p - q + r = \alpha$ (D) $p + q + r = \beta$

UD0173

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Phy\Module\1-U&D, Basic Maths & Vector, NUM, K-1D, 2D\Eng\Eng\01_Unit & Dimension, Basic Maths and Vector\01_Eng.p65

ANSWER KEY

EXERCISE # O

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	C	A	A	B	A	C	D	D	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	A	D	B	D	B	C	B	B	C
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	D	A	C	A	C	B	A	D	B	D
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	A	B	A	C	A	C	C	C	B	B
Que.	41	42	43	44	45	46	47	48	49	
Ans.	D	B,C	C,D	A,B	B	C	A-R;B-S;C-P;D-Q	A-S;B-P;C-R;D-T	A-R;B-Q;C-P	

EXERCISE # S

1. L^{-1}, ML^2T^{-2} **2.** $kg.m^2.s^{-2}.K^{-1}$ **3.** $T = a\sqrt{\frac{m}{k}}$ **4.** $\omega = K\sqrt{\frac{GM}{r^3}}$ **5.** K = 6

6. $[MLT^{-1}]$ **7.** (i) $\frac{24}{5}$; (ii) $-\frac{7}{25}$; (iii) $\frac{\sqrt{3}-1}{2\sqrt{2}}$; (iv) $\frac{\sqrt{3}-1}{2\sqrt{2}}$ **8.** (a) $\frac{2\pi}{3}\times 6.4\times 10^6 m$, (b) $\sqrt{3}\times 6.4\times 10^6$ m

9. (a) 9.95, (b) 0.99 **10.** π cm **11.** (i) 105°, (ii) 150°, (iii) 105° **12.** $\sqrt{75}$ **13.** 1 **14.** 25

15. 120 N, $40\sqrt{3}$ N **16.** 5 **17.** $\cos\alpha = \left(\frac{4}{\sqrt{61}}\right)$, $\cos\beta = \left(\frac{6}{\sqrt{61}}\right)$ $\cos\gamma = \left(\frac{3}{\sqrt{61}}\right)$, $magnitude = \sqrt{61}$

18. 3 **19.** 90 m **20.** (a) $11\hat{i}+5\hat{j}-7\hat{k}$, (b) $\cos^{-1}\left(\frac{-7}{\sqrt{195}}\right)$, (c) $\cos^{-1}\left(\frac{-20}{\sqrt{1309}}\right)$

21. $16\hat{i}-30\hat{k}, 198J$ **22.** 2 **23.** $\frac{7}{16}$ **24.** $\hat{i}+\hat{j}$, $3\hat{k}$

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	4	4	2	2	4	3	2	3	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	4	2	4	2	4	1	1	1	3	B
Que.	21	22	23							
Ans.	C	C	B							

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6
Ans.	C	3	C	1.99 to 2.01	1,2,4	A,B

node06\B0BC BD\Kota\JEE(Advanced)\Nurture\Phy\Module\1-U& D, Basic Maths & Vector; NUM, K-1D,2D\Eng\Eng\01_Unit & Dimension, Basic Maths and Vector\01_Eng.p65